

COURSE TITLE : PACKAGING TECHNIQUES
COURSE CODE : 5101
COURSE CATEGORY : B
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 55
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Basic concept of packaging.	15
2	Materials in packaging.	15
3	Types of packaging.	13
4	Applications of packaging.	12
TOTAL		55

Course outcomes:

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To understand the concept of packaging.
2	To know the miscellaneous materials in packaging.
3	To comprehend different types of packaging .
4	To understand various application levels of packaging.

MODULE I BASIC CONCEPT OF PACKAGING

1.1.0. To understand the concept of packaging.

- 1.1.1. To define packaging
- 1.1.2. To describe functions of packaging.
- 1.1.3. To explain classification of packaging.
- 1.1.4. To describe types of packaging.
- 1.1.5. To identify various markings on packages.
- 1.1.6. To list out types of tests in packages.

MODULE II MATERIALS IN PACKAGING.

2.1.0 To know the miscellaneous materials in packaging.

- 2.1.1 To identify the adhesives used in packaging.
- 2.1.2 To describe the lamination process.
- 2.1.3 To define & classify varnishing.
- 2.1.4 To distinguish various coating.
- 2.1.5 To explain about closures.

MODULE III TYPES OF PACKAGING.

3.1.0 To comprehend different types of packaging .

- 3.1.1 To describe Aerosol packaging, Thermoforming packaging, Vacuum packaging , Shrink packaging, Aseptic packaging
- 3.1.2 To state & explain injection blow methods, Extrusion blow molding, compression molding.
- 3.1.3 To describe stretch wrapping.
- 3.1.4 To select suitable packaging method depending on the product and its requirement.

MODULE IV APPLICATIONS OF PACKAGING

4.1.0 Package preparation techniques.

- 4.1.1 To illustrate and explain carton making process
- 4.1.2 To describe corrugated board, fiber board combination
- 4.1.3 To classify additive tapes.
- 4.1.4 To explain labels.
- 4.1.5 To identify basic elements of labeling & its types.

CONTENT DETAILS

MODULE I

Packaging-Introduction. Functions of package-protection, utility, communication, containment.

Factors influencing design, classification of packaging- flexible packaging, rigid packaging type, types of packaging-consumer packaging, industrial packaging, Military packaging.

Markings on package - Handling marks, Routing marks, Information marks.

Tests on package –physical test(Mechanical test, Drop test, Vibration test, Compression test, Inclined impact test, Rolling test, Drum test) Climatic test - Rain test, Sand and Dust test, Salt Spray test, Fungus Resistance test, paperboard(cobb test),chemical test-(measure of pH, Estimate of solvent, Soluble Matter).

Cushioning Materials - Functions, Properties.

MODULE II

Materials used for Packaging - wood, board, glass, plastic, metals, glass.

Carton making. Corrugated board, plastic, Fiber Board combination.

Adhesive- Types of adhesive- Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive.

Lamination- wet lamination, Dry lamination, Hot melt lamination, Extrusion lamination. Coatings- primer coating, Protective coating, Heat sealable and Self adhesive coating.

Varnishing, Types of Varnishing- Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers.

Closure, Features and functions, Types of closure- normal, vacuum, seal, pressure seal.

MODULE III

Aerosol packaging- Aerosol pack content(product and propellant),Aerosol cans, Aerosol valve,Filling methods.

Thermoforming packaging-skin packaging, shrink packaging, blister packaging.

Vacuum packaging-gas packaging(Machanical air,Replacement, Gas flushing, Compensated vacuum),Active packaging, passive atmosphere modification.

Aseptic packaging-aseptic system, aseptic filling & packaging, classification of aseptic packaging system.

Aluminum foil, Gas packaging, MAP, CAP, Vacuum packaging, Shrink packaging, Stretch Wrapping, Strip packaging.

Moulding-Injection blow molding, Extrusion blow molding, Compression molding. Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming.

Selection of suitable method depending on the product(packing of hazardous materials, cosmetics, food products etc).

MODULE IV

Introduction to Packaging machines.

Futuristic trends in packaging.

Adhesive tapes-fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes.

Labels - Basic elements of correct labeling, Purpose.

Corrugated board preparation-Materials, components (liner, flute, adhesives). Board manufacturing.

Box manufacturing-box styles, box designs, joints, closing styles.

Packaging rules and regulations. FDA, FPO, MFPO.

Reference :		
Author	Title	Publishers
IIP	Packaging technology Vol – I,II, III	IIP india
Frank Paine	Packaging design and performance	
Jhonn Briston	Advance in plastic packaging technology	